



Risk and Logistics Assignment 2

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1 Introduction

This report aims to address the problem of designing an efficient recharging infrastructure for a fleet of autonomous surveying robots operating in Antarctica. The surveying company has deployed 1072 battery-powered robots, each equipped with a drone capable of flying to a charging station when needed. Due to harsh environmental conditions and variable energy consumption, each robot has a stochastic flying range, which determines whether it can reach a charger autonomously. If a robot cannot reach a charger within its available range, a human-operated vehicle can be deployed at an additional cost to transport it to a station.

Robots are fitted with solar panels. However, unpredictable weather and operational demands make the actual need for charging probabilistic. Specifically if a robot has range r_i the probability it will require charging is given by:

$$p_i = p(r_i) = \exp(-\lambda^2(r_i - r_{min})^2)$$

Where λ is a parameter and r_{min} is the lowest possible range of the robot.

Each charging station can host multiple chargers, with a limit of m per station and at most q robots can be allocated to a single charger and will charge to its maximum range r_{max} . The cost considerations are:

- c_b : The annualised construction investment of a charging station (build cost £/station)
- c_m : The maintenance cost of a charger (£/charger)
- $c_{\tilde{c}}$: The cost of charging (£/km)
- c_h : The cost of moving the robot with insufficient range to a charger

The objective is to determine optimal locations and capacities for charging stations to ensure all robots can be reliably charged while minimising total costs.

2 Deterministic MINLP

2.1 Problem Formulation

Sets and Indices

I : Set of robots, indexed by $i \in I$

J : Set of candidate charging station locations, indexed by $j \in J$

Parameters

(x_i, y_i) : Coordinates of robot i

r_i : Deterministic flying range of robot i

The Euclidean distance between robot i located at (x_i, y_i) and station j located at (X_j, Y_j) is given by:

$$d_{ij} = \sqrt{(x_i - X_j)^2 + (y_i - Y_j)^2}$$

Cost parameters:

$c_b = 5000$: Construction cost per station

$c_m = 500$: Maintenance cost per charger

$c_{\tilde{c}} = 0.42$: Charging cost per km

$c_h = 1000$: Human transport cost

Capacity parameters:

$m = 8$: Maximum chargers per station

$q = 2$: Maximum robots per charger

$r_{max} = 175$: Maximum robot flying range

M : Maximum distance that can be travelled in the region

Decision Variables

$y_j \in \{0, 1\} \quad \forall j \in J$: 1 if station j is built, 0 otherwise

$z_{ij} \in \{0, 1\} \quad \forall i \in I, j \in J$: 1 if robot i is assigned to station j , 0 otherwise

$h_i \in \{0, 1\} \quad \forall i \in I$: 1 if robot i requires human transport, 0 otherwise

$x_j \geq 0 \quad \forall j \in J$: Number of chargers at station j

$X_j, Y_j \in \mathbb{R} \quad \forall j \in J$: Coordinates of charging station j

Additional Modelling Considerations

In the initial implementation of the MINLP model, the station location variables (X_j, Y_j) were left unbounded. This led to numerically unstable and unrealistic solutions, with station coordinates taking extreme values outside the region of interest.

To address this, upper and lower bounds were imposed on (X_j, Y_j) based on the geographical limits of the robot locations. This restriction ensured that charging stations are placed within a realistic area and significantly improved solver stability and solution quality.

Objective Function

To minimise total costs:

$$\min \sum_{j \in J} (c_b y_j + c_m x_j) + \sum_{i \in I} \sum_{j \in J} c_{\bar{c}} \sqrt{(x_i - X_j)^2 + (y_i - Y_j)^2} z_{ij} + \sum_{i \in I} c_h h_i$$

Constraints

Each robot must be assigned to exactly one station

$$\sum_{j \in J} z_{ij} = 1 \quad \forall i \in I$$

Assignment can only occur if station is built

$$z_{ij} \leq y_j \quad \forall i \in I, j \in J$$

Each charger serves at most q robots

$$\sum_{i \in I} z_{ij} \leq q x_j \quad \forall j \in J$$

Each station can have a maximum of m chargers

$$0 \leq x_j \leq m y_j$$

If a robot cannot reach the assigned station it must incur human transport cost

$$\sqrt{(x_i - X_j)^2 + (y_i - Y_j)^2} z_{ij} \leq r_i + Mh_i \quad \forall i \in I, j \in J$$

Here, the maximum distance the robot would have to travel to reach any station is bounded above by M , which is the maximum distance that can be traversed across the region of interest. This value of M ensures that the constraint is always satisfied when the robot requires human transport.

2.2 Small instance solution

The MINLP model was solved for 20 robots and 10 robots. Due to the complexity of the problem, the model reached its time limit, producing a result with a 25% MIP Gap and 1% MIP Gap for 20 and 10 robots respectively. The results are presented in Table 1.

Characteristic	20 Robots	10 Robots
Number of robots	20	10
Number of stations	2	1
Construction cost	£10,000	£5,000
Maintenance cost	£5,000	£2,500
Charging cost	£125.35	£75.31
Human transport cost	£0	£0
Total cost	£15,125.35	£7,575.31
Lower bound on Total Cost	£11,250	£7,500
MIP Gap	25%	0.99%

Table 1: Results from MINLP model for 20 robots

The results from the model provide us with an upper and lower bound for the optimal solution of both instances. In the following sections, a construction heuristic and improvement heuristic will give insight into the optimal solution for the small instance as well as the bigger problem.

3 Construction Heuristic

The goal of the problem is to determine the optimal placement of charging stations for a set of robots, accounting for construction and maintenance costs of stations and chargers, charging costs proportional to distance, capacity limits of chargers per station, range limitations of robots and human transport requirements. The heuristic approach provides a fast, feasible solution for the deterministic case where all robot ranges are known.

3.1 Methodology

Each robot is assigned a weight inversely proportional to its range, giving priority to shorter ranged robots. Weighted K-Means clustering is applied to robot coordinates to generate initial station locations. An initial number of stations is estimated as:

$$k_0 = \left\lceil \frac{N}{m \cdot q} \right\rceil$$

Stations exceeding capacity are recursively split using K-Means with $k = 2$ on the subset of robots assigned to the overloaded station. This process repeats until all stations respect capacity

constraints.

Each robot is assigned to the nearest station after splitting. Chargers are allocated to each station proportional to the number of assigned robots, capped at m . Robots beyond their range to assigned stations are flagged for human transport.

The total cost is computed as:

$$\min \sum_{j \in J} (c_b y_j + c_m x_j) + \sum_{i \in I} \sum_{j \in J} c_{\bar{c}} d_{ij} z_{ij} + \sum_{i \in I} c_h h_i$$

3.2 Solving for Small Instance

The construction heuristic was applied to the instance of finding station coordinates for 20 robots. The results are presented in Table 2.

Characteristic	Value
Number of stations	2
Number of chargers	10
Construction cost	£10,000
Maintenance cost	£5,000
Charging cost	£82.73
Human transport cost	£0
Total cost	£15,082.73

Table 2: Characteristics of the final solution for 20 robots obtained using the construction heuristic.

3.3 Results

The construction heuristic produced a feasible initial solution with 107 charging stations, ensuring that all robots could be assigned to a station within their flying range without requiring human transport.

The total cost of this initial solution was approximately £800,000. While feasible, the solution is relatively expensive due to the high number of stations, leading to significant construction and maintenance costs. This highlights a need for further refinement, motivating the use of a local search heuristic to reduce redundancy in station placement.

The results have been summarised in Table 3.

Characteristic	Value
Number of stations	107
Number of chargers	560
Construction cost	£535,000
Maintenance cost	£280,000
Charging cost	£981.68
Human transport cost	£0
Total cost	£815,981.68

Table 3: Characteristics of the full final solution obtained using the construction heuristic.

3.4 Performance Measures and Guarantees

The performance of the deterministic construction heuristic was evaluated using several measures including total cost, cost breakdown, number of stations built, number of chargers installed, number of robots transported by humans, and computational time.

Performance Guarantees

The construction heuristic provides the following guarantees:

1. **Feasibility Guarantee:** The heuristic always produces a feasible solution by definition of the heuristic where every robot is assigned to a station and station capacities are respected.
2. **Theoretical Lower Bound:** The absolute minimum possible cost is calculated by finding the minimum number of stations required for the robots we have.

$$LB = c_b \cdot \left\lceil \frac{N}{m \cdot q} \right\rceil$$

For the 20-robot instance: $LB = 5000 \times \lceil 20/16 \rceil = 10,000$ The heuristic achieved 15,082.73, giving a gap of 50.8% from this weak bound.

3. **Empirical Performance:** When compared to the MINLP solution (which had a 25% MIP gap), the construction heuristic produced costs within 0.3% of the MINLP upper bound for the 20-robot instance in a smaller duration

Performance Summary Table

Table 4 contains our summary of performance results for the construction heuristic.

Metric	20 Robots	1072 Robots
Total Cost	£15,082.73	£800,000 (approx.)
Number of Stations	2	107
Number of Chargers	10	N/A
Human Transport	0%	0%
Computation Time	0.04 seconds	1.65 seconds
Gap from Theoretical LB	50.8%	N/A

Table 4: Construction heuristic performance across instance sizes

4 Local Search Heuristic

The construction heuristic produced a feasible solution where every robot could fly to a charging station. However, the solution contained many charging stations which leads to high construction and maintenance costs. If a station only serves a small number of robots it may be cheaper to reassign those robots to nearby stations and close that station down, saving the stations fixed costs. The local search heuristic therefore attempts to improve the construction solution by eliminating stations and reallocating their robots to other stations when this reduces the total cost.

4.1 Neighbourhood Structure

The neighbourhood is defined by solutions obtained by removing a single charging station and reassigning its robots to other stations. For a reassignment to be feasible, two conditions must be satisfied:

Each robot must be able to reach its new station:

$$d_{ij} \leq r_i$$

Station capacity must not be exceeded. The maximum number of robots per station is:

$$m \cdot q = 8 \cdot 2 = 16$$

4.2 Station Elimination

Stations are considered for elimination starting from the smallest stations (defined as those serving the fewest robots), since these are more likely to be redundant.

For a candidate station s , the algorithm first identifies all alternative stations that each robot assigned to s can reach within its range. If a robot has no feasible alternative station, the elimination attempt is immediately rejected.

If every robot has at least one candidate station, the algorithm attempts to reassign all robots from station s to other stations while respecting capacity constraints. Since multiple robots may share the same candidate stations, a backtracking search procedure is used to find a feasible assignment.

If a feasible reassignment is found, the algorithm evaluates the resulting change in total cost. The cost difference includes:

- Savings from removing the construction cost of station s ,
- Savings from removing the maintenance cost of chargers at station s ,
- Changes in charging cost due to different travel distances,
- Additional maintenance costs if extra chargers are required at the receiving stations.

The elimination move is only accepted if it reduces the total cost.

4.3 Station Re-centering

After the elimination phase, an additional improvement step is performed. For each station j , the centroid of the locations of the robots assigned to that station is computed. Moving the station to this centroid can potentially reduce the total flying distance and therefore the charging cost. The station location is only updated if all robots assigned to the station remain within their flying range and the total cost decreases.

4.4 Solving for Small Instance

The local search heuristic was applied to the instance of finding station coordinates for 20 robots. This heuristic had a £0.12 improvement in total costs when compared to the construction heuristic. The results are presented in Table 5.

Characteristic	Value
Number of stations	2
Number of chargers	10
Construction cost	£10,000
Maintenance cost	£5,000
Charging cost	£82.60
Human transport cost	£0
Total cost	£15,082.60

Table 5: Characteristics of the final solution for 20 robots obtained using the local search heuristic.

4.5 Results

The local search heuristic was applied to the solution obtained from the construction heuristic. Starting from an initial solution with 107 stations, the algorithm successfully removed redundant stations by reassigning robots to nearby stations while maintaining feasibility.

The elimination phase prioritised smaller stations, which are more likely to be cost inefficient. After the elimination phase, the number of stations was reduced significantly from 107 to 68 with a total cost of £610,198.58. This heuristic had an improvement of £205,783.10 when compared to the construction heuristic.

The re-centering step produced small improvements by relocating several stations slightly closer to the centroid of their assigned robots. This reduced the total charging distance and therefore the charging cost.

Table 6 contains the final solution characteristics.

Characteristic	Value
Number of stations	68
Number of chargers	537
Construction cost	£340,000
Maintenance cost	£268,500
Charging cost	£2,212.80
Human transport cost	£0
Total cost	£610,198.58

Table 6: Characteristics of the final solution obtained using the local search heuristic.

All constraints are satisfied in the final solution. In particular, no station exceeds its capacity and all robots are within flying range of their assigned charging station.

4.6 Performance Measures and Guarantees

The deterministic local search heuristic improves the solution by relocating stations and reassigning robots, reducing travel distances and overall cost. Table 7 contains our small instance performance results.

Performance on Small Instance (20 Robots)

Metric	Construction	Local Search	Improvement
Total Cost	£15,082.73	£15,082.60	£0.13 (0.0009%)
Number of Stations	2	2	0
Charging Cost	£82.73	£82.60	£0.13
Human Transport	0	0	-

Table 7: Local search improvement on 20-robot instance

The small improvement on the 20-robot instance indicates the construction heuristic already produced a near-optimal solution for this small problem. Table 8 contains our full instance results.

Performance on Full Instance (1072 Robots)

Metric	Construction	Local Search	Improvement
Total Cost	£815,981.68	£610,198.58	£205,783.10 (25.2%)
Number of Stations	107	68	39 (36.4% reduction)
Number of Chargers	N/A	537	-
Construction Cost	£535,000	£340,000	£195,000
Maintenance Cost	N/A	£268,500	-
Charging Cost	N/A	£2,212.80	-

Table 8: Local search improvement on full 1072-robot instance

Computational Time Efficiency Metrics

- Construction heuristic: 1.65 seconds
- Local search heuristic: 1.65 seconds
- Total: 3.3 seconds for 1072 robots

Solution Quality Guarantees

While local search does not guarantee global optimality, it provides:

1. **Local Optimality:** The final solution is locally optimal with respect to station elimination (no single station can be removed without increasing cost).
2. **Feasibility Maintenance:** All constraints remain satisfied throughout the search.
3. **Monotonic Improvement:** Cost strictly decreases with each accepted move.
4. **Termination Guarantee:** The algorithm terminates after a finite number of iterations (no improvement after 10 consecutive attempts).

Comparison with MINLP Lower Bound

For the 20-robot instance, the MINLP solver provided a lower bound of £11,250 and an upper bound of £15,125.35 (25% gap). Both heuristics produced solutions within 0.3% of the MINLP upper bound, validating their effectiveness for small instances.

For the full instance, no optimal solution exists computationally, but the 25.2% improvement from local search demonstrates the value of the improvement heuristic.

Comparison with 10 robots case

While applying the heuristics on the case of 10 robots, it was seen that the heuristic could construct a feasible solution in a shorter time span than that of the MIP Solver, but the objective value of the function was significantly higher. It is seen that the heuristics provided a better solution than the MIP solver solution for the case of 20 robots in comparison to the case of 10.

These measures indicate that the heuristic results in a better solution when the number of robots considered are higher while for lower instances, the MIP solver attains a reasonably good feasible solution for a longer time span.

5 Stochastic Problem Formulation

5.1 Two Stage Stochastic Program

In the deterministic setting each robot has a fixed known range r_i . However, in real life scenarios, the range of each robot is uncertain and can depend on several things such as weather conditions and variable energy consumption from operations. This uncertainty is modelled explicitly using $S = 100$ scenarios, where in scenario s robot i has range r_i .

We model this as a two-stage stochastic programme using Sample Average Approximation (SAA). In the first stage we decide where to build charging stations and how many chargers to install at each before the scenario is known. In the second stage we decide which robots fly to their assigned station and which require human transport. These recourse decisions adapt to the realised ranges.

The robots do not need charging every day. If robot i has range r_i^s in scenario s , the probability that it will require charging is:

$$p_i = \exp(-\lambda^2(r_i - r_{min})^2)$$

Robots with a range close to r_{min} are almost certain to need charging, while robots with a high range rarely do. Each robot's contribution to the variable cost is therefore weighted by p_i^s .

5.2 Problem Formulation

The model is extended from the deterministic version of the problem. We define the following modifications for the stochastic problem.

Sets: S - Set of scenarios

Parameters:

- r_i^s - range of robot i in scenario s
- p_i^s - probability robot i will charge in scenario s

- p_s - probability of scenario s , i.e., $\frac{1}{|S|}$ as all scenarios are assumed to be equally likely

Decision Variables:

1. Stage 1:

- $y_j \in \{0, 1\}$ - binary variable to decide if station j is built or not
- X_j, Y_j - continuous variable for determining coordinates of the stations
- $x_j \in \mathbb{Z}^+$ - number of chargers at station j
- $z_{ij} \in \{0, 1\}$ - binary variable indicating if robot i is assigned to station j

2. Stage 2:

- h_i^s - a binary variable indicating if robot i requires human transport for scenario s

Constraints: All constraints from the deterministic model remain the same except the reachability constraint. This is because the range of the robot is dependent on the scenario and thus the choice of station locations is modelled such that under any scenario, the station the robot is allocated to is reachable for the robot, either by flying in its range or using human transport. Thus, the constraint is modified to:

$$\sqrt{(x_i - X_j)^2 + (y_i - Y_j)^2} z_{ij} \leq r_i^s + M h_i^s \quad \forall i \in I, j \in J, s \in S$$

Objective Function:

We aim to minimise the expected cost of the station openings. The cost of opening a station and the maintenance cost per charger are fixed. Thus, they are calculated as before. We have that the cost for charging is dependent on the range of the robot as well as the probability that it will charge. The expected cost of charging the robots is given as

$$\text{Expected Charging Cost} = \sum_{s \in S} p_s \sum_{i \in I} \sum_{j \in J} p_i^s (c_{\bar{c}} \cdot d_{ij} \cdot (1 - h_i^s) + c_h \cdot h_i^s)$$

Therefore,

$$\text{Total Expected Cost} = \sum_{j \in J} (c_b y_j + c_m x_j) + \sum_{s \in S} p_s \sum_{i \in I} \sum_{j \in J} p_i^s (c_{\bar{c}} \cdot d_{ij} \cdot (1 - h_i^s) + c_h \cdot h_i^s)$$

Thus, the objective function is

$$\min \underbrace{\sum_{j \in J} (c_b y_j + c_m x_j)}_{\text{Stage 1 Fixed Costs}} + \underbrace{\sum_{s \in S} p_s \sum_{i \in I} \sum_{j \in J} p_i^s (c_{\bar{c}} \cdot d_{ij} \cdot (1 - h_i^s) + c_h \cdot h_i^s)}_{\text{Stage 2 Expected Costs}}$$

5.3 Algorithm

Since solving the full two-stage stochastic programme is computationally expensive for 1072 robots and 100 scenarios, we develop a heuristic solution approach combining a construction heuristic with a stochastic local search procedure.

Firstly, the weighted K-Means construction heuristic from the previous section is used to obtain an initial feasible station layout. The weight for each robot is calculated as the inverse of the average charging range across all scenarios, where the robots with smaller expected ranges

have greater influence on station placement. The number of initial stations is chosen based on the total charging capacity, and overloaded stations are iteratively split using weighted K-Means clustering until all capacity constraints are satisfied.

Using the solution from the construction heuristic as the initial solution, a stochastic local search heuristic is then applied. The local search attempts to eliminate stations and reassign their robots to nearby stations while not violating capacity and deterministic range feasibility constraints. Candidate moves are evaluated using the expected cost over all scenarios.

The expected cost is computed by averaging charging and human transport costs across all scenarios, weighted by the probability that each robot requires charging under each scenario. By fixing the set of scenarios and optimising the expected cost over them, the heuristic produces a solution that approximates the optimal solution to the stochastic facility location problem.

5.4 Results

Table 9 compares the deterministic and stochastic solutions.

Characteristic	Deterministic	Stochastic
Number of stations	68	68
Construction cost	£340,000	£340,000
Maintenance cost	£268,500	£268,500
Charging cost	£2,212.80	£3,945.52
Human transport cost	£0	£30,771.01
Total cost	£610,198.58	£638,072.97

Table 9: Characteristics of the final solution obtained using the Stochastic model compared to the Deterministic model

The station layout remains unchanged suggesting that the deterministic solution is already robust to range uncertainty such that structural changes would not be beneficial.

The total cost is approximately 5% higher in the stochastic case, representing the cost of ignoring uncertainty when designing the infrastructure.

Human transport is the main source of additional costs as the deterministic case assumes perfect information. For some scenarios the robots realised range falls short of the distance to its assigned station.

In the deterministic model, every robot is assumed to charge with probability 1, inflating the expected charging cost. The stochastic model correctly weights each robot’s charging cost by its probability of needing charge. Since stochastic charging cost is higher despite this weighting, it is understood that the station configuration itself requires longer average flight distances.

6 Conclusion

This report investigated the design of cost-efficient charging infrastructure for a fleet of autonomous robots operating under two conditions, deterministic and stochastic. A Mixed-Integer Non-Linear Programming model was developed to provide an exact formulation of the problem. The model produced high-quality solutions for small instances of 20 robots with a 25% MIP gap. However, due to the computational complexity of the problem, this model was impractical for larger scale problems. Hence, motivating the development of heuristics.

A construction heuristic was formulated based on weighted K-Means clustering to generate feasible initial station locations. This approach ensured feasibility, but it resulted in an excessive number of charging stations and consequently high fixed costs. To address this, a local search

heuristic was developed, focusing on eliminating redundant stations and reassigning robots while maintaining feasibility. This significantly reduced the number of stations and total cost.

The model was then extended to a stochastic setting to account for uncertainty in robot ranges and charging requirements. Using a Sample Average Approximation approach, the stochastic solution showed that the deterministic design is relatively robust as the optimal station configuration remained unchanged. Incorporating uncertainty did lead to an increase in total costs, particularly due to human transport costs in scenarios where robots could not reach charging stations.

Overall, the results highlight the importance of balancing infrastructure investment with operational flexibility. While deterministic models can provide useful baseline solutions, ignoring uncertainty can underestimate real world costs. The proposed heuristic framework offers a practical and scalable approach for large instances, producing near optimal and robust solutions within reasonable computation times.